



WELLSHINE DISTRIBUTORS

Technical Operations & Architecture Manual

Project Scope:	Premium Food Wholesaling Portal (Cashews, Dry Fruits, Spices)
Domain Name:	wellshinedistributors.com
System Engine:	Node.js / Express.js / PostgreSQL
Deployment:	Render (Web Server) & Supabase (Database Hosting)

1. Executive & Technical Summary

Wellshine Distributors is a premium wholesale food supplier based in Pala, Kottayam, Kerala. The web portal functions as a high-performance business portfolio and ordering gateway. It bridges the gap between digital local search ranking (SEO) and transactional business-to-business (B2B) bulk commerce.

This manual details the languages, database functionality, backend schemas, search-engine indexing architecture, and deployment infrastructures configured to run the system safely and efficiently.

2. Core Technology Stack

Programming Languages & Environment

- **Node.js (Runtime):** Powers the high-performance backend, serving API routing, database pooling, session verifications, and email services.
- **JavaScript (ES6+):** Used on both server-side (asynchronous route controllers) and client-side (dynamic catalogs, interactive shopping carts, and UI animations).
- **HTML5 & CSS3:** Structured pages using semantic HTML elements and responsive styling utilizing custom design tokens (light/dark variables) and glassmorphism.

Frameworks, Libraries & Dependencies

- **Express.js (Backend Framework):** Acts as the web controller, managing routing, static directories, CORS headers, security middlewares, and HTTPS redirects.
 - **pg (node-postgres):** Maintains active connection pools to PostgreSQL, compiling queries and managing parameter binding securely.
 - **JSON Web Token (JWT):** Enforces encrypted stateless user session payloads, signing and verifying customer credentials.
 - **bcryptjs:** Hashes user registration passwords securely on the server using salt-rounds before storage in the database.
 - **Nodemailer:** Integrates SMTP services to trigger automatic customer OTP verification emails and checkout notices.
 - **Jimp:** Automates server-side logo and image processing, optimizing dimensions and compiling favicons.
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3. Database Engine & Schema Layout

The application is powered by PostgreSQL, a robust relational SQL database. The active production database is hosted on Supabase, which guarantees ACID-compliant operations, automatic backups, and low-latency scaling.

Core SQL Schema Design

Table: products

Stores the prioritized catalog. Configured with stock levels and trigger-volume discount rules.

- **id (INT, PRIMARY KEY):** Unique identifier for each product.
- **name (VARCHAR), price (INT), unit (VARCHAR):** Product catalog details.
- **tag (VARCHAR), cat (VARCHAR), img (VARCHAR):** Category classification and asset URL.
- **description (TEXT):** Detailed descriptions optimized for search crawler reading.
- **stock (INT), discount_percent (INT), discount_trigger_qty (INT):** Inventory details and B2B pricing triggers.

Table: users

Stores registered B2B clients and credentials.

- **id (INT, PRIMARY KEY):** Unique identifier.
- **email (VARCHAR, UNIQUE):** Verified customer email address.
- **password_hash (VARCHAR):** Cryptographically secure bcrypt password hash.
- **shop_name (VARCHAR), address (TEXT), pincode (VARCHAR):** Store billing and delivery coordinates.
- **account_type (VARCHAR):** Client tier designation (e.g., Business, Wholesale).

Table: orders

Maintains customer order queues.

- **id (INT, PRIMARY KEY):** Unique order serial identifier.
- **user_id (INT):** Foreign key linking back to the verified customer.
- **items (TEXT/JSON):** Serialized snapshot of catalog items, quantities, and prices.
- **total_amount (INT):** Net invoice amount in Rupees.
- **status (VARCHAR):** Order tracking status (e.g., Pending, Confirmed, Shipped).
- **created_at (TIMESTAMP):** Auto-generated order timestamp.

4. Application Functionality & Security

Core Operations & Workflows

- **Server-Side Pre-rendering (SSR):** When a crawler or visitor opens /products, the Express server queries PostgreSQL, builds the product card elements, and sends a complete HTML page. This guarantees instant indexing by search engine crawlers.
- **Customer Checkout Flow:** Registered customers log in using secure OTP checks, add products to their shopping cart, and submit orders. The server creates an order record and triggers email notifications to both the customer and admin.
- **Automatic B2B Quantity Discounts:** The system computes discounts dynamically. For example, W320 Cashews automatically apply a 5% discount when ordering above the trigger threshold, calculated on the fly on both client and server.
- **Admin Portal:** A secure cockpit allowing administrative users to edit product information, manage stock levels, track orders, change fulfillment statuses, and manage user tiers.

5. System Security & Compliance

The platform is hardened using a multi-layer security configuration to protect customer data and database integrity:

- **Row-Level Security (RLS) on Supabase:** Row-Level Security is enabled on all tables. This completely blocks direct anonymous API access via the Supabase HTTP gateway from the internet. Only queries originating from your backend superuser pooler (which bypasses RLS securely) are processed.
 - **Bcryptjs Hashing:** No plain text passwords are ever stored. Passwords are salted and hashed on the server, protecting credentials against database exposure.
 - **JWT Sessions:** Stateless JSON Web Tokens are signed using a cryptographically secure server secret. Customer tokens are validated on every private API transaction.
 - **SQL Injection Prevention:** All database calls utilize parameterized SQL query binding. Variables are passed separately from query strings, neutralizing database injection attacks.
 - **HTTPS Enforced Redirection:** In production, Express detects insecure connections and automatically redirects all traffic to secure HTTPS, encrypting session payloads in transit.
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6. Search Engine Optimization (SEO)

The website has been optimized to rank highly for local search queries (e.g. "cashews in Pala", "dryfruits in Kottayam") through several key integrations:

- **LocalBusiness JSON-LD Schema:** Embedded metadata telling search crawlers the physical coordinates, phone number, Kattakayam Road address, and operating hours of Wellshine Distributors.
- **FAQPage Schema & Accordions:** An interactive, glassmorphic FAQ accordion on the homepage answers target queries, coupled with JSON-LD FAQPage schemas to trigger rich search result accordion snippets on Google.
- **Clean Extensionless URLs:** Express rewrites routes, allowing URLs like /products and /shop to load without exposing .html file extensions.
- **Optimized Favicon Array:** Includes explicit high-resolution sizes (192x192 PNG logo) and standard favicon.ico fallbacks to ensure correct Google search results favicon indexing.

7. Deployment, Domain & Environment

Domain Details

- **Registered Domain:** wellshinedistributors.com
- **DNS SSL/TLS:** Managed securely, enforcing automatic secure HTTPS connection handshakes.

Web Server Hosting (Render)

- **Host Provider:** Render Cloud Platform
- **Operating Environment:** Linux VM executing Node.js (V8 runtime engine) in a production setup.
- **Process Management:** Auto-restarting server instances connected to the master GitHub repository branch for continuous deployment.

Database Hosting (Supabase)

- **Host Provider:** Supabase Cloud PostgreSQL Hosting (Project Reference: asjnaimlzlqrtcncfwk).
 - **Connection Mode:** PostgreSQL connection pooling via Transaction Mode (Port 6543) using a secure DATABASE_URL.
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